

SANDEEP KUMAR

M.Tech, Fuel and Energy Engineering — Computational Modeling — Python — Energy Systems

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Professional Summary

M.Tech candidate at IIT (ISM) Dhanbad with experience in computational modeling (CFD, finite element analysis), Python programming, and technical literature research on energy and industrial processes. Seeking an Analyst (Data & Research) role to apply quantitative analysis and modeling to energy-sector problems.

Education

Indian Institute of Technology (Indian School of Mines) Dhanbad

Expected May 2027

M.Tech, Fuel and Energy Engineering (Dept. of Fuel, Minerals and Metallurgical Engineering) — **CGPA: 6.4/10** Dhanbad, Jharkhand

- **Relevant Coursework:** Computational Fluid Dynamics, Processing of Liquid and Gaseous Fuels, Biofuels, Alternate Energy Systems, Powerplant Engineering

Rajiv Gandhi College of Engineering, Research and Technology, Chandrapur

August 2021

Bachelor of Engineering in Mining Engineering — **CGPA: 9.88/10**

Chandrapur, Maharashtra

- **Relevant Coursework:** Mine Ventilation, Mine Surveying, Mine Planning, Rock Mechanics, Mine Legislation, Mining Machinery

Projects

Criterion for the Onset of Air Core in a Hydrocyclone

2026 – Present

- Reviewing literature on swirl flow and air core behavior to identify the research gap in predicting onset conditions.
- Developing a CFD model in OpenFOAM and ANSYS Fluent (VOF multiphase, RSM turbulence) and planning lab experiments with high-speed imaging to validate a dimensionless onset criterion.

Stress and Deformation Analysis in High-Temperature Extrusion of a Special Material for Nuclear Applications

Jun 2026 – Jul 2026

- Built coupled structural and thermal finite element simulations in ANSYS to quantify extrusion force, stress, strain, and temperature distribution on the extrusion die under operating conditions.
- Identified high-stress and high-temperature zones prone to wear, cracking, and premature die failure, providing data-driven inputs for die design, material selection, and cooling strategy to extend die life and reduce downtime and replacement costs.

Training and Internship

Nuclear Fuel Complex

Jun 2026 – Jul 2026

Intern

Hyderabad, Telangana

- Completed a one-month technical internship across the major processing facilities at NFC Hyderabad, gaining exposure to nuclear fuel fabrication and metallurgical manufacturing workflows.
- Studied Extrusion and Piercing Plant operations, including thermo-mechanical processing, high-pressure metal working, and heavy-equipment mechanics.
- Examined process parameters, precision engineering protocols, and quality-control measures for seamless tube and rod manufacturing.

Jhanjra Project Colliery (Eastern Coalfields Limited)

Jan 2022 – Jan 2023

Post Graduation Practical Training (PGPT)

Jhanjra, West Bengal

- Observed production workflows and daily operations at a high-capacity underground coal mine.
- Studied the mine ventilation system and its role in airflow planning, worker safety, and mine planning.
- Monitored the deployment and cycle times of heavy-duty underground machinery to understand its effect on productivity.

Manganese Ore India Limited (MOIL)

Dec 2019 – Jan 2020

Vocational Trainee

Ukwa, Balaghat, Madhya Pradesh

- Observed the opening of a new shaft and the operational planning involved in developing underground access.
- Used survey instruments and safety devices in fieldwork and studied the mining methods used in a metal mine.

Technical Skills

- **Programming & Tools:** Python, C++, MATLAB, SQL, MS-Excel
- **Simulation & Modeling:** ANSYS Fluent, OpenFOAM, ANSYS Structural and Thermal (FEA), Life Cycle Assessment (LCA)
- **Core Engineering Areas:** Thermodynamics, Computational Fluid Dynamics (CFD), Heat Transfer, Fluid Mechanics, Energy Systems